

Database Healthcare Assistance Program Finder – ER Modeling

Note: During Relational Schema and SQL phase it was discovered that user is reserved name, the name was changed to ApplicantUser

1. Requirement Analysis

For each user, the system stores a unique user ID, first name, last name, ZIP code, annual income, insurance type, annual out-of-pocket healthcare expenses, phone number, and email.

For each organization, the system stores a unique organization ID, name, type, city, state, phone number, email, and website.

For each program, the system stores a unique program ID, program name, description, eligibility criteria, coverage type, and the geographic location (city and state) where the program is available.

Each organization provides one or more programs, and each program is offered by exactly one organization.

Each user can submit zero or more applications for assistance programs.

Each application has a unique application ID, application date, and application status.

Each application is submitted by exactly one user.

Each application is associated with exactly one program, and each program can receive multiple applications.

2. Conceptual Design (ER Diagram Description)

Entities

ApplicantUser

user_id

first_name

last_name

zip_code

annual_income

insurance_type

annual_out_of_pocket_expenses *this is more important than total spending for our analysis*

cell_phone

email

Organization

org_id

name

type

city

state

phone

email

website

Program

prog_id

program_name *e.g. Cancer Aid Fund*

description *e.g. Helps cover hospital and treatment costs for cancer patient*

eligibility_criteria *e.g age of >65 (for mediCare), low income, resident of certain state*

coverage_type *e.g. Vision, Emergency, Primary care*

city

state

Application

app_id

app_date

app_status *e.g. submitted, under review, approved, rejected*

Relationships

Offers: Organization to Program (1:N, total participation on both sides)

Submits: ApplicantUser to Application (1:N, total participation on Application side)

Receives: Program to Application (1:N, total participation on Application side)

3. Assumptions

- Each organization must provide at least one program.
- Each program must be offered by exactly one organization.
- Each application must be submitted by exactly one user.
- Each application must be associated with exactly one program.
- A ApplicantUser may submit zero or many applications.
- A program may receive zero or many applications.

4. ER Diagram

